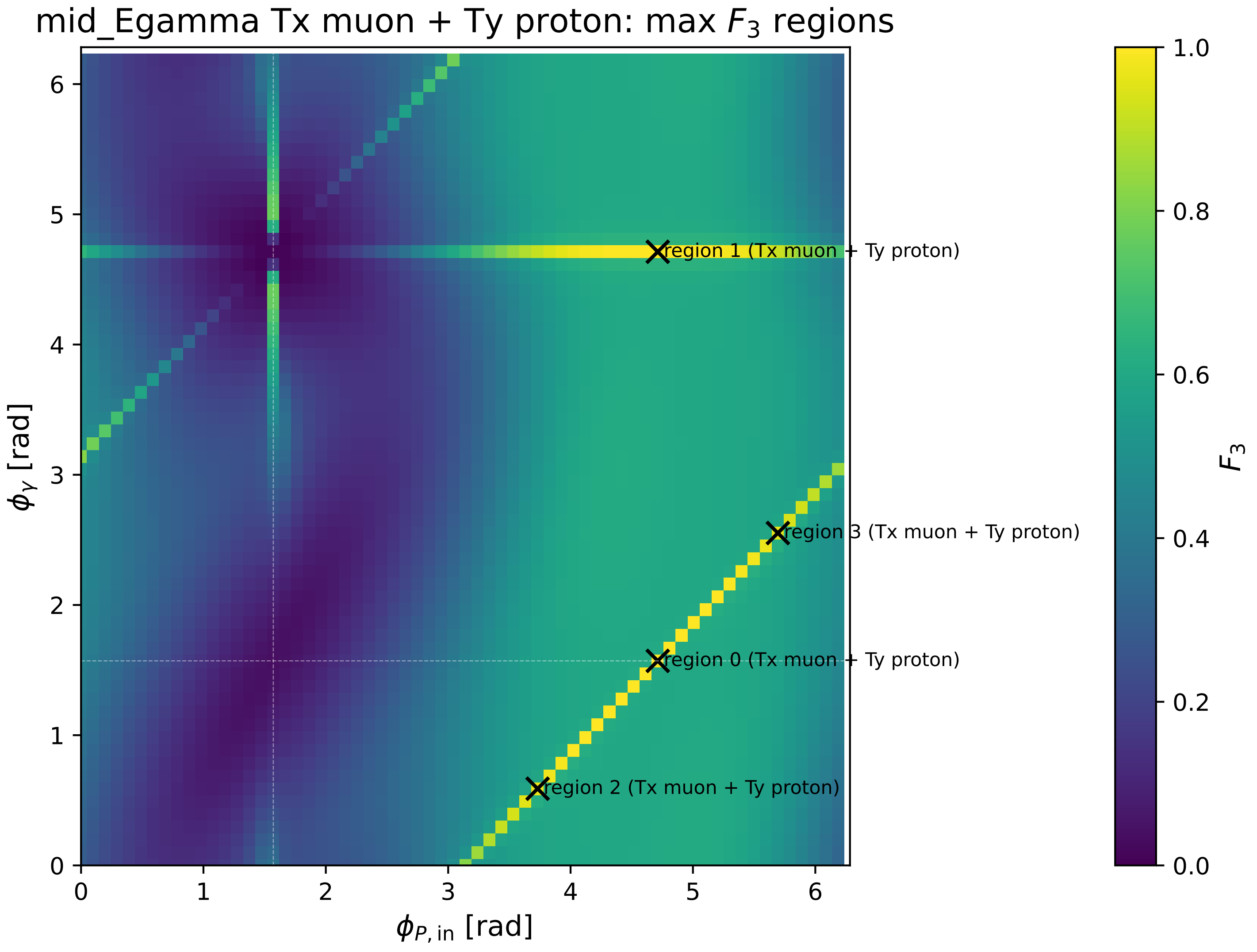
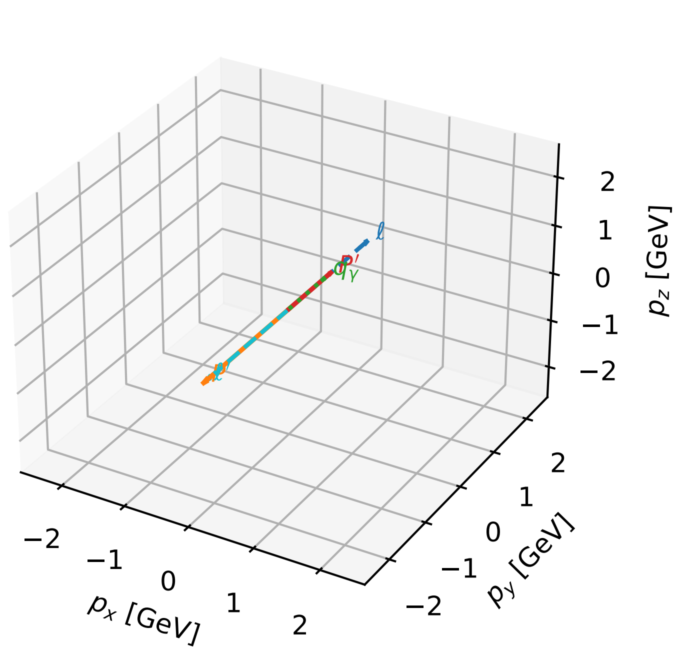


mid_Egamma Tx muon + Ty proton: max F_3 regions
region 1 (Tx muon + Ty proton)
region 3 (Tx muon + Ty proton)
region 0 (Tx muon + Ty proton)
region 2 (Tx muon + Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
0
1
2
3
4
5
6
F_3
0.0
0.2
0.4
0.6
0.8
1.0

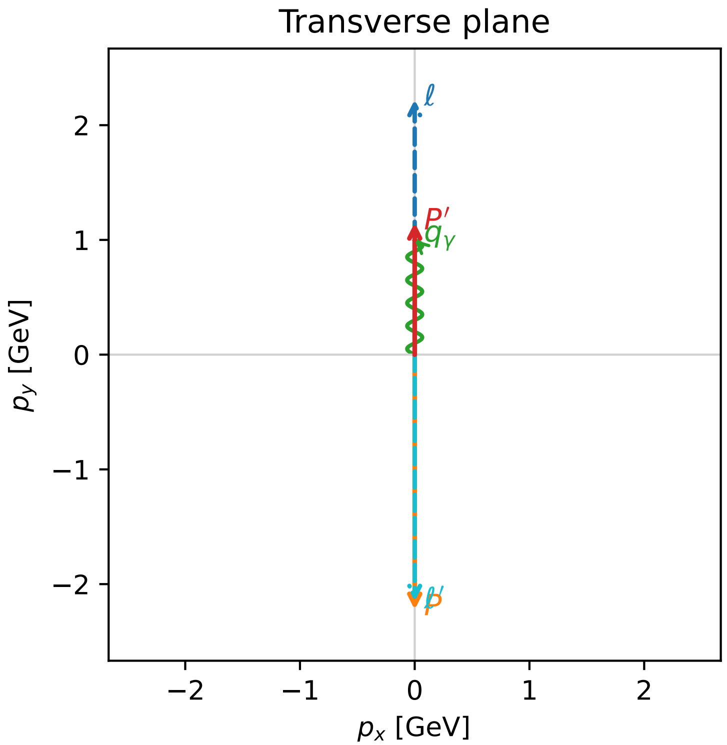
Tx muon + Ty proton characteristic max F_3 configuration

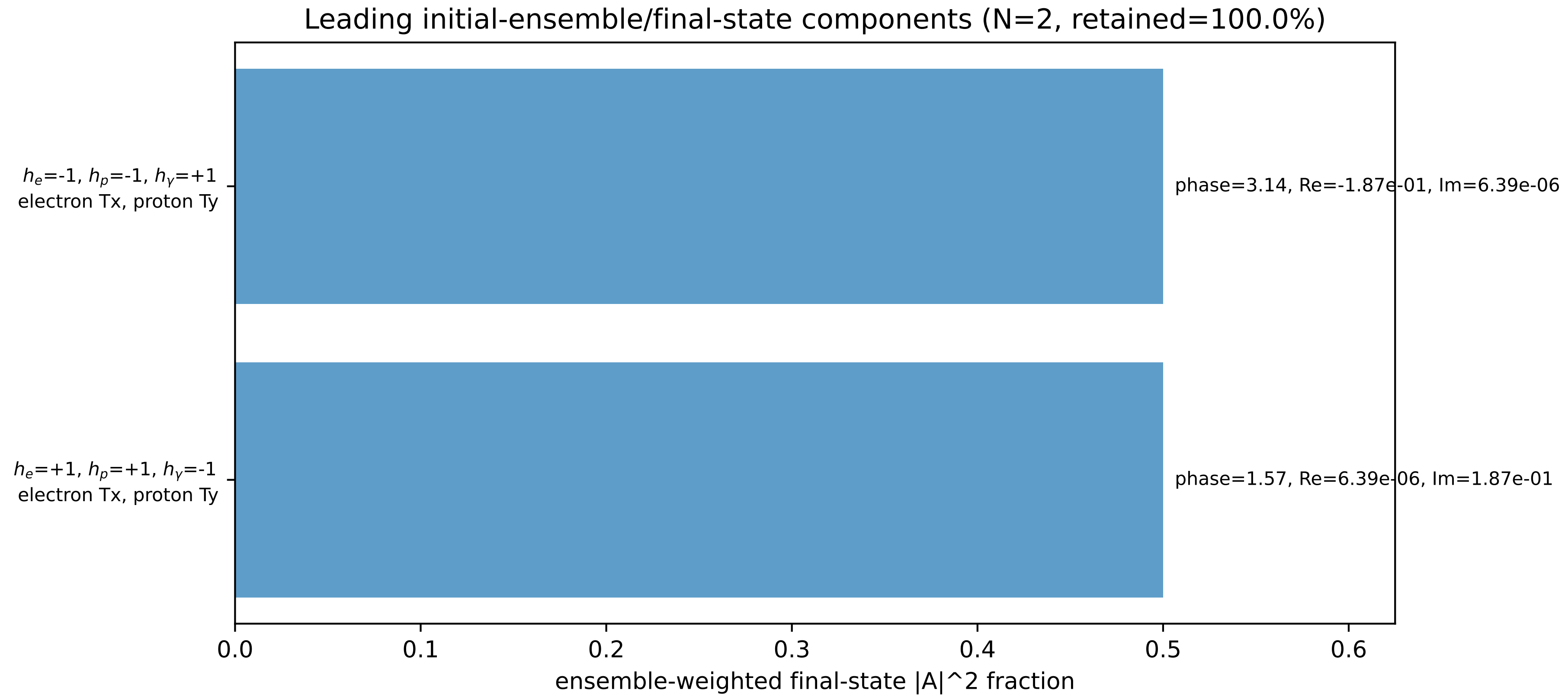


f3_mid_egamma_txty_region_0 F_3=1
region: mid_Egamma_TxTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$

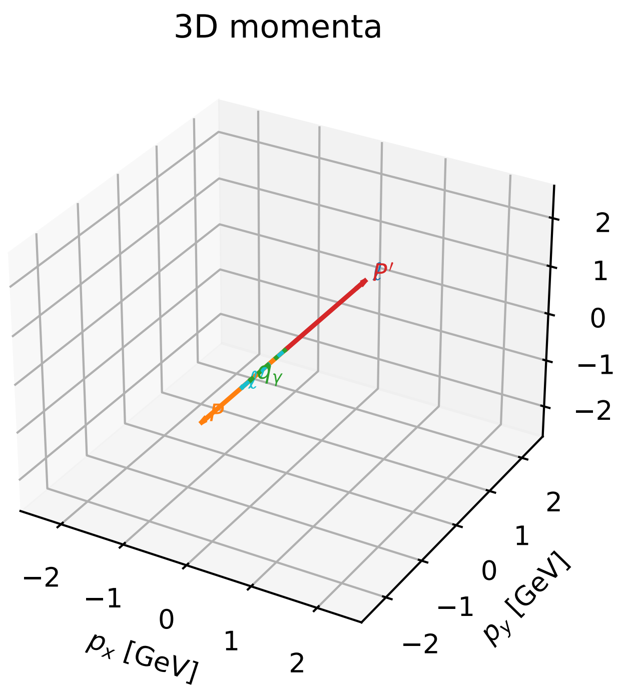
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.15107
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=1.5708
Q^2=19.1283, x_B=0.966277, t=-10.5238, W^2=1.54742, y=0.959806
$\theta(\ell', \gamma)$=180 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.0000 p_y= 2.2231 p_z= -0.0000
P E= 2.4129 p_x= -0.0000 p_y= -2.2231 p_z= 0.0000
ℓ' E= 2.1537 p_x= -0.0000 p_y= -2.1511 p_z= 0.0000
P' E= 1.4849 p_x= 0.0000 p_y= 1.1511 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.0000 p_y= 1.0000 p_z= 0.0000





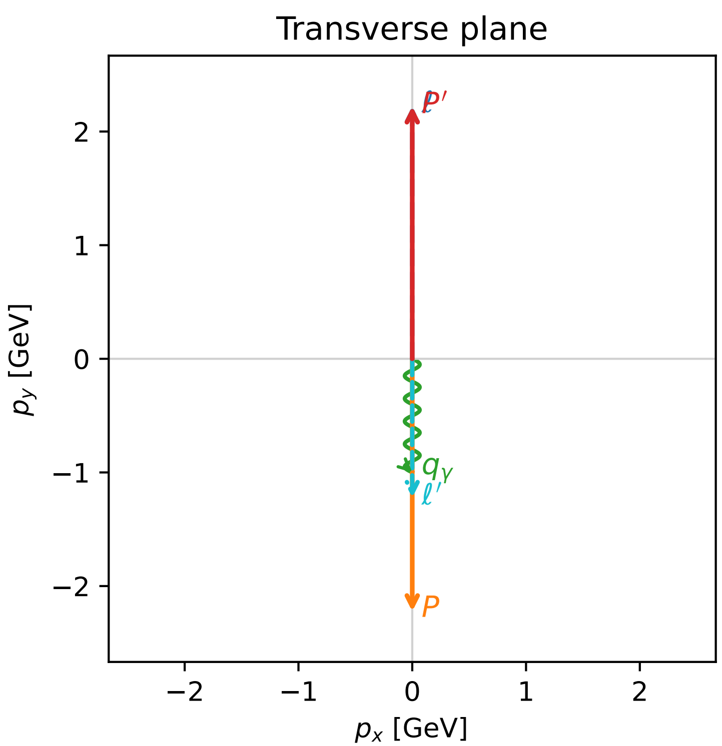
Tx muon + Ty proton characteristic max F_3 configuration

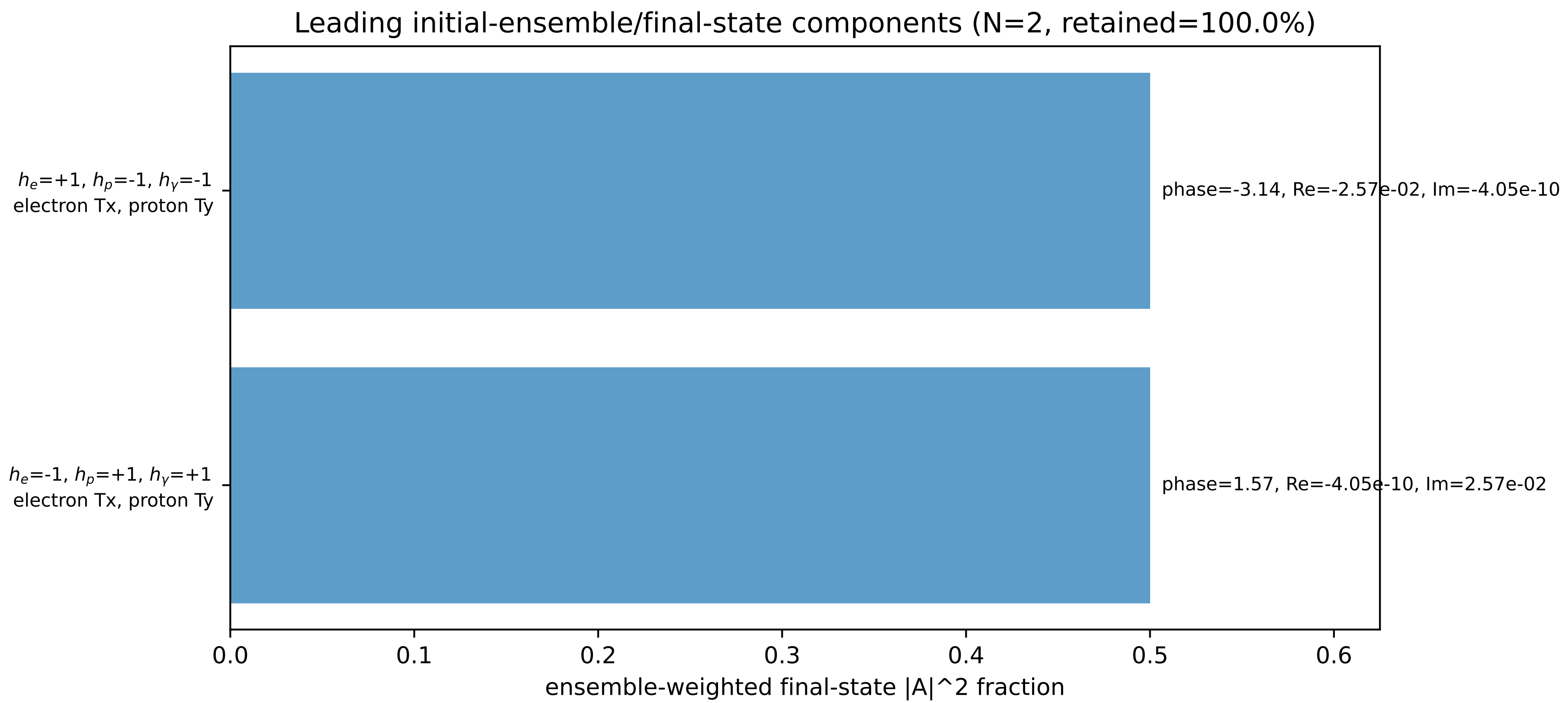


f3_mid_egamma_txty_region_1 F_3=0.999999
region: mid_Egamma_TxTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$

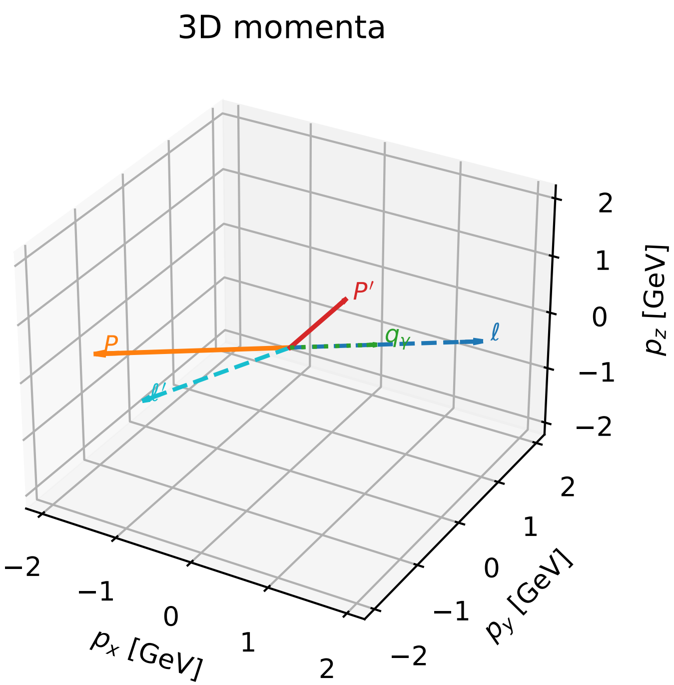
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.22204
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=4.71239
Q^2=10.8711, x_B=0.539802, t=-19.7594, W^2=10.1478, y=0.976443
$\theta(\ell', \gamma)$=0 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.0000 p_y= 2.2231 p_z= -0.0000
P E= 2.4129 p_x= -0.0000 p_y= -2.2231 p_z= 0.0000
ℓ' E= 1.2266 p_x= 0.0000 p_y= -1.2220 p_z= 0.0000
P' E= 2.4119 p_x= 0.0000 p_y= 2.2220 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.0000 p_y= -1.0000 p_z= 0.0000





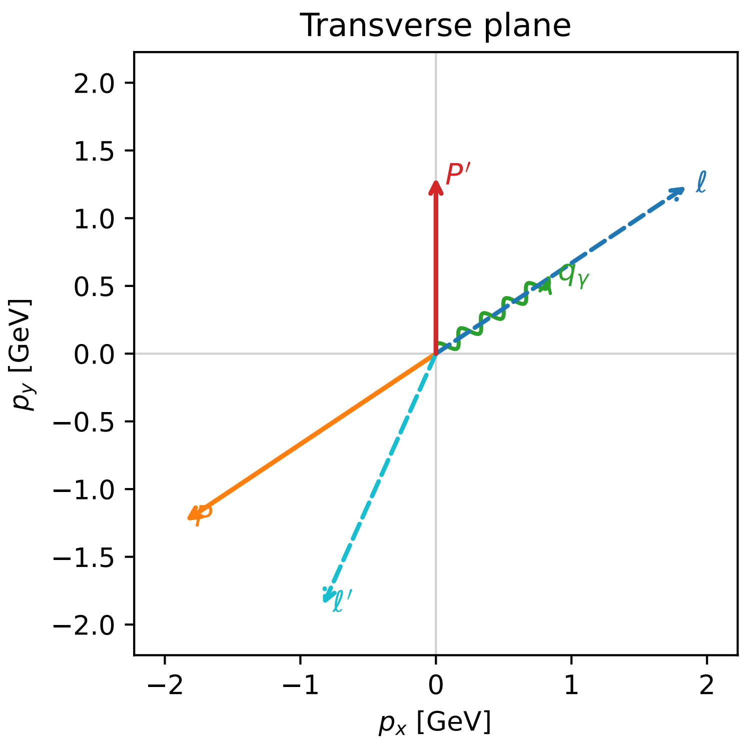
Tx muon + Ty proton characteristic max F_3 configuration

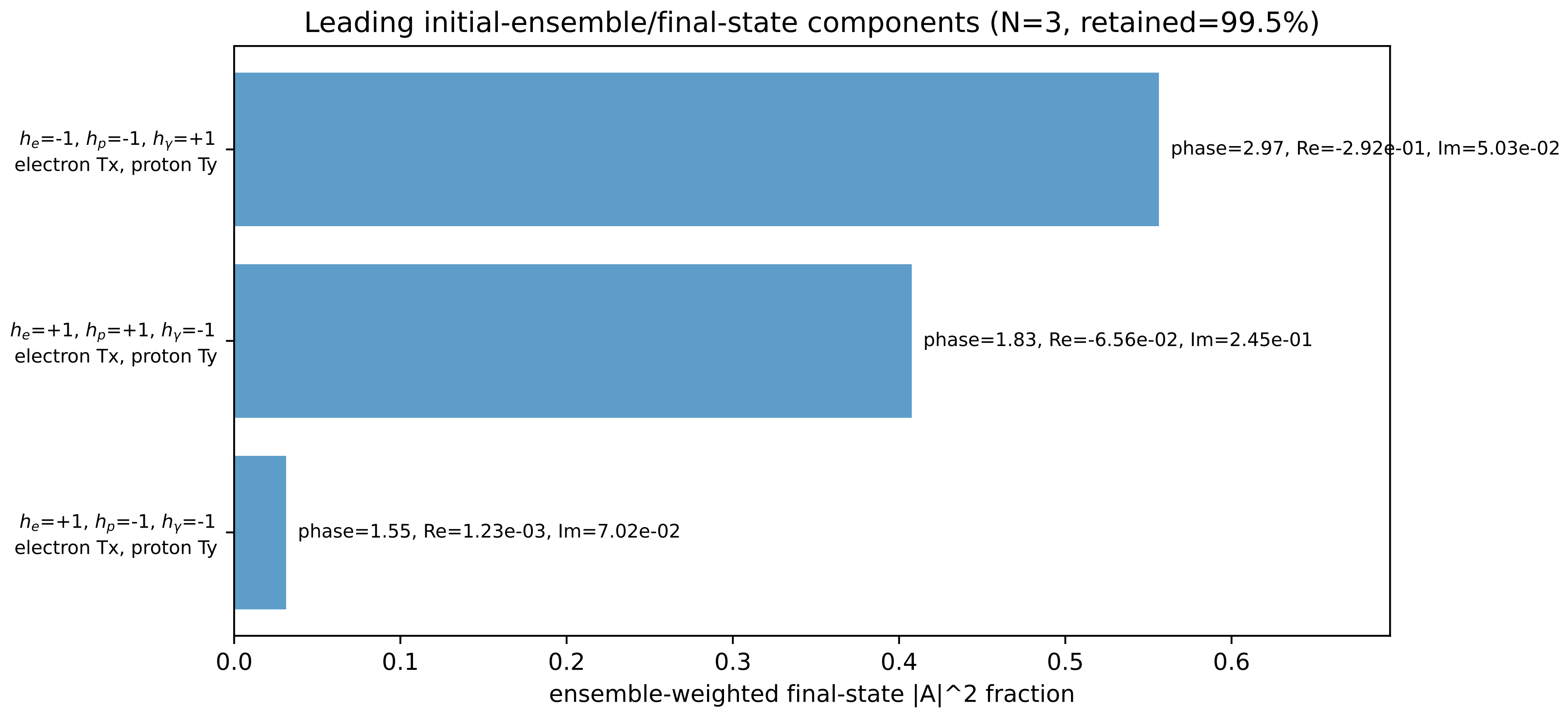


f3_mid_egamma_txty_region_2 F_3=0.968347
region: mid_Egamma_TxTy_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.29963
θ_{in}=1.57079, ϕ_ℓ=0.589049, ϕ_P=3.73064
E_γ=1, ϕ_γ=0.589049
Q^2=16.6958, x_B=0.904565, t=-9.18528, W^2=2.64132, y=0.894906
$\theta(\ell',\gamma)$=147.891 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.8484 p_y= 1.2351 p_z= -0.0000
P E= 2.4129 p_x= -1.8484 p_y= -1.2351 p_z= 0.0000
ℓ' E= 2.0357 p_x= -0.8315 p_y= -1.8552 p_z= 0.0000
P' E= 1.6028 p_x= 0.0000 p_y= 1.2996 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.8315 p_y= 0.5556 p_z= 0.0000





Tx muon + Ty proton characteristic max F_3 configuration